

# SOLIDWORKS API (.NET VBA C#) — Reference

## (123/337): IFeatureStatistics- IFillPatternFeatureData

SOLIDWORKS API Documentation (converted from CHM)

# IFeatureStatistics (Type)

## IFeatureStatistics Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFeatureStatistics Interface

Allows access to the feature statistics in a part document.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IFeatureStatistics
```

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics
```

C#

```
public interface IFeatureStatistics
```

C++/CLI

```
public interface class IFeatureStatistics
```

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### Accessors

[IFeatureManager::FeatureStatistics](#)

### Access Diagram

[FeatureStatistics](#)

### See Also

[IFeatureStatistics Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

## IFeatureStatistics Interface Methods

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFeatureStatistics Interface

For a list of all members of this type, see [IFeatureStatistics members](#).

### Public Methods

| Name                    | Description                                          |
|-------------------------|------------------------------------------------------|
| <a href="#">Refresh</a> | Refreshes the feature statistics in a part document. |

[Top](#)

## See Also

[IFeatureStatistics Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IFeatureStatistics Interface Members

[See Also](#) [Properties](#) [Methods](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFeatureStatistics Interface

The following tables list the members exposed by [IFeatureStatistics](#).

### Public Properties

| Name                                                                                                                           | Description                                                                                                                                                                 |
|--------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">FeatureCount</a>                 | Gets the number of features in a part document.                                                                                                                             |
|  <a href="#">FeatureNames</a>                 | Gets the names of the features in a part document.                                                                                                                          |
|  <a href="#">Features</a>                     | Gets the features in a part document.                                                                                                                                       |
|  <a href="#">FeatureTypes</a>                 | Gets the types of features in a part document.                                                                                                                              |
|  <a href="#">FeatureUpdatePercentageTimes</a> | Gets the percentages of time it takes to update (rebuild) each feature in a part document relative to the total time it takes to update all features in that part document. |
|  <a href="#">FeatureUpdateTimes</a>           | Gets the times, in seconds, it takes to update (rebuild) each feature in a part document.                                                                                   |
|  <a href="#">PartName</a>                     | Gets the name of the part document whose feature statistics you want.                                                                                                       |
|  <a href="#">SolidBodiesCount</a>             | Gets the number of solid bodies in a part document.                                                                                                                         |
|  <a href="#">SurfaceBodiesCount</a>           | Gets the number of surface bodies in a part document.                                                                                                                       |
|  <a href="#">TotalRebuildTime</a>             | Gets the time, in seconds, it takes to rebuild (update) all features in a part document.                                                                                    |

[Top](#)

### Public Methods

| Name                                                                                                        | Description                                          |
|-------------------------------------------------------------------------------------------------------------|------------------------------------------------------|
|  <a href="#">Refresh</a> | Refreshes the feature statistics in a part document. |

[Top](#)

## See Also

[IFeatureStatistics Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments](#) on this topic.

## IFeatureStatistics Interface Properties

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFeatureStatistics Interface

For a list of all members of this type, see [IFeatureStatistics members](#).

### Public Properties

| Name                                                                                                                             | Description                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">FeatureCount</a>                 | Gets the number of features in a part document.                                                                                                                             |
|  <a href="#">FeatureNames</a>                 | Gets the names of the features in a part document.                                                                                                                          |
|  <a href="#">Features</a>                     | Gets the features in a part document.                                                                                                                                       |
|  <a href="#">FeatureTypes</a>                 | Gets the types of features in a part document.                                                                                                                              |
|  <a href="#">FeatureUpdatePercentageTimes</a> | Gets the percentages of time it takes to update (rebuild) each feature in a part document relative to the total time it takes to update all features in that part document. |
|  <a href="#">FeatureUpdateTimes</a>           | Gets the times, in seconds, it takes to update (rebuild) each feature in a part document.                                                                                   |
|  <a href="#">PartName</a>                     | Gets the name of the part document whose feature statistics you want.                                                                                                       |
| <a href="#">SolidBodiesCount</a>                                                                                                 | Gets the number of solid bodies in a part document.                                                                                                                         |



[SurfaceBodiesCount](#)

Gets the number of surface bodies in a part document.



[TotalRebuildTime](#)

Gets the time, in seconds, it takes to rebuild (update) all features in a part document.

[Top](#)

## See Also

[IFeatureStatistics Interface](#)

[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FeatureCount Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : FeatureCount Property (IFeatureStatistics)

Gets the number of features in a part document.

## .NET Syntax

Visual Basic (Declaration)

ReadOnly Property FeatureCount As System.Integer

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics
Dim value As System.Integer
```

```
value = instance.FeatureCount
```

C#

```
System.int FeatureCount {get;}
```

C++/CLI

```
property System.int FeatureCount {
    System.int get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Number of features

## Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::FeatureCount](#).

## Example

[Get Part's Feature Statistics \(VB.NET\)](#)

[Get Part's Feature Statistics \(VBA\)](#)

[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)

[IFeatureStatistics Members](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FeatureNames Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : FeatureNames Property (IFeatureStatistics)

Gets the names of the features in a part document.

## .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureNames As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics  
Dim value As System.Object  
  
value = instance.FeatureNames
```

C#

```
System.object FeatureNames {get;}
```

C++/CLI

```
property System.Object^ FeatureNames {  
    System.Object^ get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Array of the names of the features

## Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::FeatureNames](#).

## Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## Features Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : Features Property (IFeatureStatistics)

Gets the features in a part document.

## .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property Features As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics  
Dim value As System.Object  
  
value = instance.Features
```

C#

```
System.Object Features {get;}
```

C++/CLI

```
property System.Object^ Features {  
    System.Object^ get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of [features](#)

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::Features](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FeatureTypes Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : FeatureTypes Property (IFeatureStatistics)

Gets the types of features in a part document.

### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureTypes As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics  
Dim value As System.Object  
  
value = instance.FeatureTypes
```

C#

```
System.Object FeatureTypes {get;}
```

C++/CLI

```
property System.Object^ FeatureTypes {  
    System.Object^ get();  
}
```

```
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of the types of features as defined by [swSelectType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics.FeatureTypes](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FeatureUpdatePercentageTimes Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : FeatureUpdatePercentageTimes Property (IFeatureStatistics)

Gets the percentages of time it takes to update (rebuild) each feature in a part document relative to the total time it takes to update all features in that part document.

### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property FeatureUpdatePercentageTimes As System.Object
```

Visual Basic (Usage)

```
Dim instance As IFeatureStatistics
Dim value As System.Object

value = instance.FeatureUpdatePercentageTimes
```

C#

```
System.Object FeatureUpdatePercentageTimes {get;}
```

C++/CLI

```
property System.Object^ FeatureUpdatePercentageTimes {
    System.Object^ get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of percentages of time it takes to update (rebuild) each feature in a part document relative to the total time it takes to update all features in that part document

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics.FeatureUpdatePercentageTimes](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)  
[IFeatureStatistics::FeatureUpdateTimes Property](#)  
[IFeatureStatistics::TotalRebuildTime Property](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## FeatureUpdateTimes Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : FeatureUpdateTimes Property (IFeatureStatistics)

Gets the times, in seconds, it takes to update (rebuild) each feature in a part document.

## .NET Syntax

### Visual Basic (Declaration)

```
ReadOnly Property FeatureUpdateTimes As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IFeatureStatistics  
Dim value As System.Object
```

```
value = instance.FeatureUpdateTimes
```

### C#

```
System.Object FeatureUpdateTimes {get;}
```

### C++/CLI

```
property System.Object^ FeatureUpdateTimes {  
    System.Object^ get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Array of times, in seconds, it takes to update (rebuild) each feature in a part document

## Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::FeatureUpdateTimes](#).

## Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)  
[IFeatureStatistics::FeatureUpdatePercentageTimes Property](#)  
[IFeatureStatistics::TotalRebuildTime Property](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

## PartName Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : PartName Property (IFeatureStatistics)

Gets the name of the part document whose feature statistics you want.

### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property PartName As System.String
```

Visual Basic (Usage)



```
Dim instance As IFeatureStatistics  
Dim value As System.String  
  
value = instance.PartName
```

C#

```
System.string PartName {get;}
```

C++/CLI

```
property System.String^ PartName {  
    System.String^ get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Name of part document

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::PartName](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

## Refresh Method (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : Refresh Method (IFeatureStatistics)

Refreshes the feature statistics in a part document.

### .NET Syntax

Visual Basic (Declaration)

```
Function Refresh() As System.Boolean
```

Visual Basic (Usage)

```
Dim instance As IFeatureStatistics  
Dim value As System.Boolean  
  
value = instance.Refresh()
```

C#

```
System.bool Refresh()
```

C++/CLI

```
System.bool Refresh();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Return Value

True to refresh the feature statistics, false to not

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics.Refresh](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SolidBodiesCount Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : SolidBodiesCount Property (IFeatureStatistics)

Gets the number of solid bodies in a part document.

### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SolidBodiesCount As System.Integer
```

Visual Basic (Usage)

```
Dim instance As IFeatureStatistics  
Dim value As System.Integer  
  
value = instance.SolidBodiesCount
```

C#

```
System.int SolidBodiesCount {get;}
```

C++/CLI

```
property System.int SolidBodiesCount {
    System.int get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Number of solid bodies

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::SolidBodiesCount](#).

### Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

### See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)  
[IFeatureStatistics::SurfaceBodiesCount Property](#)

### Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SurfaceBodiesCount Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : SurfaceBodiesCount Property (IFeatureStatistics)

Gets the number of surface bodies in a part document.

### .NET Syntax

#### Visual Basic (Declaration)

```
ReadOnly Property SurfaceBodiesCount As System.Integer
```

#### Visual Basic (Usage)



```
Dim instance As IFeatureStatistics
Dim value As System.Integer
```

```
value = instance.SurfaceBodiesCount
```

#### C#

```
System.int SurfaceBodiesCount {get;}
```

#### C++/CLI

```
property System.int SurfaceBodiesCount {
    System.int get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Number of surface bodies

### Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::SurfaceBodiesCount](#).

## Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)  
[IFeatureStatistics::SolidBodiesCount Property](#)

## Availability

SOLIDWORKS 2010 FCS, Revision Number 18.0

[SOLIDWORKS API Help](#)

[Send comments](#) on this topic.

## TotalRebuildTime Property (IFeatureStatistics)

[See Also](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFeatureStatistics Interface](#) : TotalRebuildTime Property (IFeatureStatistics)

Gets the time, in seconds, it takes to rebuild (update) all features in a part document.

## .NET Syntax

### Visual Basic (Declaration)

ReadOnly Property TotalRebuildTime As System.Double

### Visual Basic (Usage)



```
Dim instance As IFeatureStatistics
Dim value As System.Double

value = instance.TotalRebuildTime
```

### C#

```
System.double TotalRebuildTime {get;}
```

### C++/CLI

```
property System.double TotalRebuildTime {
    System.double get();
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Time, in seconds, it takes to rebuild (update) all features

## Visual Basic for Applications (VBA) Syntax

See [FeatureStatistics::TotalRebuildTime](#).

## Example

[Get Part's Feature Statistics \(VB.NET\)](#)  
[Get Part's Feature Statistics \(VBA\)](#)  
[Get Part's Feature Statistics \(C#\)](#)

## See Also

[IFeatureStatistics Interface](#)  
[IFeatureStatistics Members](#)  
[IFeatureStatistics::FeatureUpdatePercentageTimes Property](#)  
[IFeatureStatistics::FeatureUpdateTimes Property](#)



# IFillPatternFeatureData (Type)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IFillPatternFeatureData Interface

[See Also](#) [Members](#) [Example](#)



[SolidWorks.Interop.sldworks Namespace](#) : IFillPatternFeatureData Interface

Allows access to a fill pattern feature.

**NOTE:** Click the **Members** link, located near the top of the topic, to see this interface's methods and properties.

### .NET Syntax

Visual Basic (Declaration)

```
Public Interface IFillPatternFeatureData
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
```

C#

```
public interface IFillPatternFeatureData
```

C++/CLI

```
public interface class IFillPatternFeatureData
```

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData](#).

### Example

'VBA

'\*\*\*\*\*

'1. Ensure the specified part exists.

'2. Run the macro.

'3. Inspect the graphics area and the FeatureManager design tree.

'\*\*\*\*\*

```
Dim swApp As SldWorks.SldWorks
```

```
Dim myFeature As SldWorks.Feature
```

```
Dim swFeat As SldWorks.Feature
```

```
Dim swFeatMgr As SldWorks.FeatureManager
```

```
Dim Part As SldWorks.ModelDoc2
```

```
Dim swFeatData As SldWorks.FillPatternFeatureData
```

```
Dim boolstatus As Boolean
```

```
Dim longstatus As Long, longwarnings As Long
```

```
Option Explicit
```

```
Sub main()
```

```
    Set swApp = Application.SldWorks
```

```
    Set Part = swApp.OpenDoc6("C:\Users\Public\Documents\SOLIDWORKS\SOLIDWORKS 2019\samples\tutorial\api\block20.sldprt", 1,  
0, "", longstatus, longwarnings)
```

```
    swApp.ActivateDoc2 "block20.sldprt", False, longstatus
```

```
    Set Part = swApp.ActiveDoc
```

```
    boolstatus = Part.Extension.SelectByRay(-0.044420257315096, 3.96239999998329E-02, -3.92581286577638E-02,  
-0.400036026779312, -0.515038074910024, -0.758094294050284, 1.1224765174324E-03, 2, False, 0, 0)
```

```
    Set myFeature = Part.FeatureManager.SimpleHole2(0.0054, True, False, False, 1, 0, 0.0254, 0.0254, False, False, False,  
1.74532925199433E-02, 1.74532925199433E-02, False, False, False, True, True, True, True, False)
```

```
    Part.SelectionManager.EnableContourSelection = False
```

```
    Part.ActivateSelectedFeature
```

```
    boolstatus = Part.Extension.SelectByID2("Hole1", "BODYFEATURE", 0, 0, 0, False, 4, Nothing, 0)
```

```
    boolstatus = Part.Extension.SelectByRay(1.06119575008097E-03, 0.039624, 4.93456023787655E-02, -0.400036026779312,  
-0.515038074910024, -0.758094294050284, 1.1224765174324E-03, 1, True, 1, 0)
```

```
    boolstatus = Part.Extension.SelectByRay(6.36836165047043E-02, 3.96239999998329E-02, -6.33171793538168E-04,  
-0.400036026779312, -0.515038074910024, -0.758094294050284, 1.1224765174324E-03, 2, True, 16384, 0)
```

```

Set swFeatMgr = Part.FeatureManager

Set swFeatData = swFeatMgr.CreateDefinition(swFmFillPattern)
swFeatData.CreateSeedCutPolygonSides = 0
swFeatData.CreateSeedCutType = 0
swFeatData.Diagonal = 0.008
swFeatData.Diameter = 0.008
swFeatData.Dimension = 5.65685424949238E-03
swFeatData.FeaturesToPatternType = 0
swFeatData.GeometryPattern = False
swFeatData.InnerRadius = 3.80422606518061E-03
swFeatData.InstanceSpacing = 1.70240490494193E-02
swFeatData.LayoutSpacingType = 1
swFeatData.LoopSpacing = 1.70240490494193E-02
swFeatData.Margins = 0
swFeatData.NoOfInstances = 10
swFeatData.OuterRadius = 0.004
swFeatData.PatternLayoutPolygonSides = 0
swFeatData.PatternLayoutType = 0
swFeatData.Rotation = 0
swFeatData.SeedCutFlipShapeDirection = False
swFeatData.StaggerAngle = 1.0471975511966
Set swFeat = swFeatMgr.CreateFeature(swFeatData)

End Sub

```

## Example

[Create Fill Pattern \(C#\)](#)

## Remarks

Read [Pattern Features and their Feature Data Objects](#).

Before calling `IFeatureManager::CreateDefinition`, you must pre-select the entities needed to create the fill pattern feature.

| Use <code>ModelDocExtension::SelectByID2</code> to select... | For the Fill Pattern's...                                                                                                                         | Using Mark... |
|--------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| Sketch, face, planar curve on a face, or coplanar faces      | Fill boundary                                                                                                                                     | 16384         |
| Direction reference (e.g., edge or axis)                     | Pattern direction (if not selected, SOLIDWORKS chooses a direction reference appropriate for the specified layout)                                | 1             |
| Vertex or sketch point                                       | Location of the pattern seed instance (if a point is not selected, the pattern seed instance is placed at the center of the fill boundary region) | 3             |
| Features                                                     | Pattern objects                                                                                                                                   | 4             |
| Faces                                                        | Pattern objects                                                                                                                                   | 5             |
| Bodies                                                       | Pattern objects                                                                                                                                   | 6             |

After calling `IFeatureManager::CreateDefinition`, you can explicitly set other properties on the feature data object. You can also initialize:

- [IFillPatternFeatureData::PatternLayoutType](#)
- [IFillPatternFeatureData::LayoutSpacingType](#)
- [IFillPatternFeatureData::FeaturesToPatternType](#)
- [IFillPatternFeatureData::GeometryPattern](#)
- [IFillPatternFeatureData::LoopSpacing](#) (for all `PatternLayoutTypes` except `swPatternLayoutType_e.swPatternLayoutPerforation`)
- Depending on the settings above, other properties might need to be initialized.

You must call [IFeatureManager::CreateFeature](#) after initializing this feature data object in order to successfully create the fill pattern feature.

For all pattern layouts except `swPatternLayoutType_e.swPatternLayoutPerforation`, the fill pattern is created by positioning a pattern seed instance within the fill boundary and copying the pattern in concentric loops about the seed instance.

See the **Fill Pattern PropertyManager** topic in the SOLIDWORKS Help for more information about fill pattern features.

## Accessors

[IFeature::GetDefinition](#)

[IFeatureManager::CreateDefinition](#)

## Access Diagram

[FillPatternFeatureData](#)

## See Also

[IFillPatternFeatureData Members](#)  
[SolidWorks.Interop.sldworks Namespace](#)

# IFillPatternFeatureData Interface Methods

[See Also](#)

[SolidWorks.Interop.sldworks Namespace](#) : IFillPatternFeatureData Interface

For a list of all members of this type, see [IFillPatternFeatureData members](#).

## Public Methods

| Name                                                                                                                     | Description                                                              |
|--------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------|
|  <a href="#">AccessSelections</a>       | Gains access to the selections used to define this fill pattern feature. |
|  <a href="#">ReleaseSelectionAccess</a> | Releases access to selections that define this fill pattern feature.     |

[Top](#)

## See Also

[IFillPatternFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

# IFillPatternFeatureData Interface Members

[See Also](#) [Properties](#) [Methods](#)

[SolidWorks.Interop.sldworks Namespace](#) : IFillPatternFeatureData Interface

The following tables list the members exposed by [IFillPatternFeatureData](#).

## Public Properties

| Name                                                                                                                          | Description                                                                                                                                       |
|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">AutoSelect</a>                 | Gets or sets whether to automatically select all or only specific bodies in a multibody part to be affected by this fill pattern feature.         |
|  <a href="#">CreateSeedCutPolygonSides</a> | Gets or sets the number of sides in a polygon-cut seed instance of this fill pattern feature.                                                     |
|  <a href="#">CreateSeedCutType</a>         | Gets or sets the type of cut for this fill pattern's seed instance.                                                                               |
|  <a href="#">Diagonal</a>                  | Gets or sets the length of the diagonal of a diamond-cut seed instance of this fill pattern feature.                                              |
|  <a href="#">Diameter</a>                  | Gets or sets the diameter of a circle-cut seed instance of this fill pattern feature.                                                             |
|  <a href="#">Dimension</a>                 | Gets or sets the length of a side of a diamond-cut or square-cut seed instance of this fill pattern feature.                                      |
|  <a href="#">FeatureScope</a>              | Gets or sets whether to use scope for this fill pattern feature in a multibody part.                                                              |
|  <a href="#">FeatureScopeBodies</a>        | Gets or sets the solid bodies that the fill pattern feature affects in a multibody part.                                                          |
|  <a href="#">FeaturesToPatternType</a>     | Gets or sets the type of object to pattern in this fill pattern feature.                                                                          |
|  <a href="#">GeometryPattern</a>           | Gets or sets whether to create this fill pattern using an exact copy of the seed feature.                                                         |
|  <a href="#">InnerRadius</a>               | Gets or sets the radius of the circle that inscribes the polygon-cut seed instance of this fill pattern feature.                                  |
|  <a href="#">InstanceSpacing</a>           | Gets or sets the distance between the pattern instance centers of this fill pattern feature.                                                      |
|  <a href="#">LayoutSpacingType</a>         | Gets or sets the type of spacing between the instances in the layout of this fill pattern feature.                                                |
|  <a href="#">LoopSpacing</a>               | Gets or sets the distance between loops of pattern instances in this fill pattern feature.                                                        |
|  <a href="#">Margins</a>                   | Gets or sets the distance between the fill boundary and the outermost pattern instance in the layout of this fill pattern feature.                |
|  <a href="#">NoOfInstances</a>             | Gets or sets the number of instances per loop or side of the layout of this fill pattern feature.                                                 |
|  <a href="#">OuterRadius</a>               | Gets or sets the radius of the circle that circumscribes the polygon-cut seed instance of this fill pattern feature.                              |
|  <a href="#">PatternBodyArray</a>          | Gets or sets the array of bodies to pattern in this fill pattern feature.                                                                         |
|  <a href="#">PatternDirection</a>          | Gets or sets the direction reference for the layout of this fill pattern feature.                                                                 |
|  <a href="#">PatternDirectionType</a>      | Gets the type of pattern direction reference returned by <a href="#">IFillPatternFeatureData::PatternDirection</a> for this fill pattern feature. |
|  <a href="#">PatternElement</a>            | Gets or sets the pattern element selection of this fill pattern feature.                                                                          |
|  <a href="#">PatternFaceArray</a>          | Gets or sets the array of faces to pattern in this fill pattern feature.                                                                          |
|  <a href="#">PatternFeatureArray</a>       | Gets or sets the array of features to pattern in this fill pattern feature.                                                                       |

|                                                                                   |                                           |                                                                                                                                                                      |
|-----------------------------------------------------------------------------------|-------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">PatternFillBoundaryArray</a>  | Gets or sets the array of objects representing the boundary of this fill pattern feature.                                                                            |
|  | <a href="#">PatternLayoutPolygonSides</a> | Gets or sets the number of sides in the polygonal layout of this fill pattern feature.                                                                               |
|  | <a href="#">PatternLayoutType</a>         | Gets or sets the layout of the pattern instances within the fill boundary of this fill pattern feature.                                                              |
|  | <a href="#">PropagateVisualProperty</a>   | Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all fill pattern instances in this fill pattern feature. |
|  | <a href="#">Rotation</a>                  | Gets or sets the counterclockwise rotation of the seed instance of this fill pattern feature.                                                                        |
|  | <a href="#">SeedCutFlipShapeDirection</a> | Gets or sets whether to reverse the direction of the seed instance cut of this fill pattern feature.                                                                 |
|  | <a href="#">SeedFeatureCenter</a>         | Gets or sets the vertex or sketch point where to place the center of the seed cut feature of this fill pattern feature.                                              |
|  | <a href="#">SeedFeatureCenterType</a>     | Gets or sets the type of entity where to place the center of the seed cut feature for this fill pattern feature.                                                     |
|  | <a href="#">SkippedItemArray</a>          | Gets or sets the pattern instances to skip in this fill pattern feature.                                                                                             |
|  | <a href="#">StaggerAngle</a>              | Gets or sets the angle by which to stagger the rows of pattern instances in the perforation layout of this fill pattern feature.                                     |

[Top](#)

## Public Methods

| Name                                                                              | Description                                                                                                    |
|-----------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|
|  | <a href="#">AccessSelections</a><br>Gains access to the selections used to define this fill pattern feature.   |
|  | <a href="#">ReleaseSelectionAccess</a><br>Releases access to selections that define this fill pattern feature. |

[Top](#)

## See Also

[IFillPatternFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## IFillPatternFeatureData Interface Properties

[See Also](#)

 [SolidWorks.Interop.sldworks Namespace](#) : IFillPatternFeatureData Interface

For a list of all members of this type, see [IFillPatternFeatureData members](#).

## Public Properties

| Name                                                                                | Description                                                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | <a href="#">AutoSelect</a><br>Gets or sets whether to automatically select all or only specific bodies in a multibody part to be affected by this fill pattern feature. |
|  | <a href="#">CreateSeedCutPolygonSides</a><br>Gets or sets the number of sides in a polygon-cut seed instance of this fill pattern feature.                              |
|  | <a href="#">CreateSeedCutType</a><br>Gets or sets the type of cut for this fill pattern's seed instance.                                                                |
|  | <a href="#">Diagonal</a><br>Gets or sets the length of the diagonal of a diamond-cut seed instance of this fill pattern feature.                                        |
|  | <a href="#">Diameter</a><br>Gets or sets the diameter of a circle-cut seed instance of this fill pattern feature.                                                       |
|  | <a href="#">Dimension</a><br>Gets or sets the length of a side of a diamond-cut or square-cut seed instance of this fill pattern feature.                               |
|  | <a href="#">FeatureScope</a><br>Gets or sets whether to use scope for this fill pattern feature in a multibody part.                                                    |
|  | <a href="#">FeatureScopeBodies</a><br>Gets or sets the solid bodies that the fill pattern feature affects in a multibody part.                                          |
|  | <a href="#">FeaturesToPatternType</a><br>Gets or sets the type of object to pattern in this fill pattern feature.                                                       |
|  | <a href="#">GeometryPattern</a><br>Gets or sets whether to create this fill pattern using an exact copy of the seed feature.                                            |
|  | <a href="#">InnerRadius</a><br>Gets or sets the radius of the circle that inscribes the polygon-cut seed instance of this fill pattern feature.                         |
|  | <a href="#">InstanceSpacing</a><br>Gets or sets the distance between the pattern instance centers of this fill pattern feature.                                         |
|  | <a href="#">LayoutSpacingType</a><br>Gets or sets the type of spacing between the instances in the layout of this fill pattern feature.                                 |
|  | <a href="#">LoopSpacing</a><br>Gets or sets the distance between loops of pattern instances in this fill pattern feature.                                               |
|  | <a href="#">Margins</a><br>Gets or sets the distance between the fill boundary and the outermost pattern instance in the layout of this fill pattern feature.           |
|  | <a href="#">NoOfInstances</a><br>Gets or sets the number of instances per loop or side of the layout of this fill pattern feature.                                      |

|                                                                                                                             |                                                                                                                                                                      |
|-----------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <a href="#">OuterRadius</a>                | Gets or sets the radius of the circle that circumscribes the polygon-cut seed instance of this fill pattern feature.                                                 |
|  <a href="#">PatternBodyArray</a>          | Gets or sets the array of bodies to pattern in this fill pattern feature.                                                                                            |
|  <a href="#">PatternDirection</a>          | Gets or sets the direction reference for the layout of this fill pattern feature.                                                                                    |
|  <a href="#">PatternDirectionType</a>      | Gets the type of pattern direction reference returned by <a href="#">IFillPatternFeatureData.PatternDirection</a> for this fill pattern feature.                     |
|  <a href="#">PatternElement</a>            | Gets or sets the pattern element selection of this fill pattern feature.                                                                                             |
|  <a href="#">PatternFaceArray</a>          | Gets or sets the array of faces to pattern in this fill pattern feature.                                                                                             |
|  <a href="#">PatternFeatureArray</a>       | Gets or sets the array of features to pattern in this fill pattern feature.                                                                                          |
|  <a href="#">PatternFillBoundaryArray</a>  | Gets or sets the array of objects representing the boundary of this fill pattern feature.                                                                            |
|  <a href="#">PatternLayoutPolygonSides</a> | Gets or sets the number of sides in the polygonal layout of this fill pattern feature.                                                                               |
|  <a href="#">PatternLayoutType</a>         | Gets or sets the layout of the pattern instances within the fill boundary of this fill pattern feature.                                                              |
|  <a href="#">PropagateVisualProperty</a>   | Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all fill pattern instances in this fill pattern feature. |
|  <a href="#">Rotation</a>                  | Gets or sets the counterclockwise rotation of the seed instance of this fill pattern feature.                                                                        |
|  <a href="#">SeedCutFlipShapeDirection</a> | Gets or sets whether to reverse the direction of the seed instance cut of this fill pattern feature.                                                                 |
|  <a href="#">SeedFeatureCenter</a>         | Gets or sets the vertex or sketch point where to place the center of the seed cut feature of this fill pattern feature.                                              |
|  <a href="#">SeedFeatureCenterType</a>     | Gets or sets the type of entity where to place the center of the seed cut feature for this fill pattern feature.                                                     |
|  <a href="#">SkippedItemArray</a>          | Gets or sets the pattern instances to skip in this fill pattern feature.                                                                                             |
|  <a href="#">StaggerAngle</a>              | Gets or sets the angle by which to stagger the rows of pattern instances in the perforation layout of this fill pattern feature.                                     |

[Top](#)

## See Also

[IFillPatternFeatureData Interface](#)  
[SolidWorks.Interop.sldworks Namespace](#)

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AccessSelections Method (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : AccessSelections Method (IFillPatternFeatureData)

Gains access to the selections used to define this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

```
Function AccessSelections(  
    ByVal TopDoc As System.Object, _  
    ByVal Component As System.Object _  
) As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim TopDoc As System.Object  
Dim Component As System.Object  
Dim value As System.Boolean  
  
value = instance.AccessSelections(TopDoc, Component)
```

C#

```
System.bool AccessSelections(  
    System.object TopDoc,  
    System.object Component  
)
```

C++/CLI

```
System.bool AccessSelections(  
& System.Object^ TopDoc,  
& System.Object^ Component  
)
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Parameters

*TopDoc*

Top-level document (see **Remarks**)

*Component*

Component in which the feature is to be modified (see **Remarks**)

#### Return Value

True if the selections are successfully accessed, false if not

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::AccessSelections](#).

### Remarks

| To modify a feature in a... | Then...                                                                                                                                                                                                      |
|-----------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Part                        | <ul style="list-style-type: none"><li>TopDoc argument is the <a href="#">IModelDoc2</a> for the part</li><li>Component argument is NULL</li></ul>                                                            |
| Assembly                    | <ul style="list-style-type: none"><li>TopDoc is the IModelDoc2 object for the assembly</li><li>Component argument is the <a href="#">IComponent2</a> object in which the feature is to be modified</li></ul> |

This method puts the model into a rollback state to allow access to the selections that define this feature. You must use either of the following methods to restore the rollback state:

- [IFeature::ModifyDefinition](#) if you modified the feature
- [IFillPatternFeatureData::ReleaseSelectionAccess](#) if you did not

### See Also

[IFillPatternFeatureData Interface](#)

[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## AutoSelect Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : AutoSelect Property (IFillPatternFeatureData)

Gets or sets whether to automatically select all or only specific bodies in a multibody part to be affected by this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property AutoSelect As System.Boolean
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Boolean
```

```
instance.AutoSelect = value
```

```
value = instance.AutoSelect
```

C#

```
System.bool AutoSelect {get; set;}
```

C++/CLI

```
property System.bool AutoSelect {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to automatically select all bodies, false to select specific bodies

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::AutoSelect](#).

### Remarks

This property is valid only if:

- [IFillPatternFeatureData::GeometryPattern](#) is true,
- and -
- [IFillPatternFeatureData::FeatureScope](#) is true.

If [IFillPatternFeatureData::FeatureScope](#) is set to false, then this property defaults to false.

If this property is set to false, then specify the affected bodies using [IFillPatternFeatureData::FeatureScopeBodies](#).

For more information, see the **Fill Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## CreateSeedCutPolygonSides Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : CreateSeedCutPolygonSides Property (IFillPatternFeatureData)

Gets or sets the number of sides in a polygon-cut seed instance of this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property CreateSeedCutPolygonSides As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Integer

instance.CreateSeedCutPolygonSides = value
value = instance.CreateSeedCutPolygonSides
```

C#

```
System.int CreateSeedCutPolygonSides {get; set;}
```

C++/CLI

```
property System.int CreateSeedCutPolygonSides {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Number of sides in a polygon-cut seed instance

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::CreateSeedCutPolygonSides](#).

### Example

See the [IFillPatternFeatureData](#) example.

### Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut
- [IFillPatternFeatureData::CreateSeedCutType](#) = [swCreateSeedCutType\\_e](#).swCreateSeedCutPolygon

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## CreateSeedCutType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : CreateSeedCutType Property (IFillPatternFeatureData)

Gets or sets the type of cut for this fill pattern's seed instance.

### .NET Syntax

Visual Basic (Declaration)

```
Property CreateSeedCutType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Integer  
  
instance.CreateSeedCutType = value  
  
value = instance.CreateSeedCutType
```

C#

```
System.int CreateSeedCutType {get; set;}
```

C++/CLI

```
property System.int CreateSeedCutType {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Type of seed instance cut as defined in [swCreateSeedCutType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::CreateSeedCutType](#).

## Example

See the [IFillPatternFeatureData](#) example.

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)  
[IFillPatternFeatureData::Diameter Property \(\)](#)  
[IFillPatternFeatureData::Dimension Property \(\)](#)  
[IFillPatternFeatureData::Diagonal Property \(\)](#)  
[IFillPatternFeatureData::Rotation Property \(\)](#)  
[IFillPatternFeatureData::CreateSeedCutPolygonSides Property \(\)](#)  
[IFillPatternFeatureData::InnerRadius Property \(\)](#)  
[IFillPatternFeatureData::OuterRadius Property \(\)](#)  
[IFillPatternFeatureData::SeedCutFlipShapeDirection Property \(\)](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Diagonal Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : Diagonal Property (IFillPatternFeatureData)

Gets or sets the length of the diagonal of a diamond-cut seed instance of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property Diagonal As System.Double

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Double

instance.Diagonal = value

value = instance.Diagonal
```

C#

```
System.double Diagonal {get; set;}
```

C++/CLI

```
property System.double Diagonal {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Length of the diagonal of a diamond-cut seed instance

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::Diagonal](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut
- [IFillPatternFeatureData::CreateSeedCutType](#) = [swCreateSeedCutType\\_e](#).swCreateSeedCutDiamond

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Diameter Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : Diameter Property (IFillPatternFeatureData)

Gets or sets the diameter of a circle-cut seed instance of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

```
Property Diameter As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Double
```

```
instance.Diameter = value
```

```
value = instance.Diameter
```

C#

```
System.double Diameter {get; set;}
```

C++/CLI

```
property System.double Diameter {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Diameter of a circle-cut seed instance

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::Diameter](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut
- [IFillPatternFeatureData::CreateSeedCutType](#) = [swCreateSeedCutType\\_e](#).swCreateSeedCutCircle

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Dimension Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : Dimension Property (IFillPatternFeatureData)

Gets or sets the length of a side of a diamond-cut or square-cut seed instance of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property Dimension As System.Double

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Double

instance.Dimension = value

value = instance.Dimension
```

C#

```
System.double Dimension {get; set;}
```

C++/CLI

```
property System.double Dimension {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Length of side of diamond-cut or square-cut seed instance

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::Dimension](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut
- [IFillPatternFeatureData::CreateSeedCutType](#) is set to one of the following:
  - [swCreateSeedCutType\\_e](#).swCreateSeedCutDiamond
  - [swCreateSeedCutType\\_e](#).swCreateSeedCutSquare

## See Also

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## FeatureScope Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : FeatureScope Property (IFillPatternFeatureData)

Gets or sets whether to use scope for this fill pattern feature in a multibody part.

## .NET Syntax

### Visual Basic (Declaration)

Property FeatureScope As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Boolean  
  
instance.FeatureScope = value  
value = instance.FeatureScope
```

### C#

```
System.bool FeatureScope {get; set;}
```

### C++/CLI

```
property System.bool FeatureScope {  
    System.bool get();  
    void set ( & System.bool value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True if only specified bodies in a multibody part are affected by the fill pattern feature; false if all the bodies are affected

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::FeatureScope](#).

## Remarks

This property is valid only if [IFillPatternFeatureData::GeometryPattern](#) is true.

If this property is true, you can use [IFillPatternFeatureData::AutoSelect](#) and [IFillPatternFeatureData::FeatureScopeBodies](#) to specify affected bodies.

For more information, see the **Fill Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See [Accessing Selections that Define Features](#) for additional details.

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)  
[IFillPatternFeatureData::AutoSelect Property\(\)](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

## FeatureScopeBodies Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : FeatureScopeBodies Property (IFillPatternFeatureData)

Gets or sets the solid bodies that the fill pattern feature affects in a multibody part.

### .NET Syntax

Visual Basic (Declaration)

Property FeatureScopeBodies As System.Object

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Object
```

```
instance.FeatureScopeBodies = value
```

```
value = instance.FeatureScopeBodies
```

C#

```
System.object FeatureScopeBodies {get; set;}
```

C++/CLI

```
property System.Object^ FeatureScopeBodies {
    System.Object^ get();
    void set ( & System.Object^ value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of [bodies](#) that the fill pattern feature affects

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::FeatureScopeBodies](#).

### Remarks

This property is valid only if:

- [IFillPatternFeatureData::GeometryPattern](#) is true,
- [IFillPatternFeatureData::FeatureScope](#) is true,
- and -
- [IFillPatternFeatureData::AutoSelect](#) is false.

You cannot edit this property to null or Nothing after the feature is created with feature scope bodies. If you try, [IFeature::ModifyDefinition](#) returns true but this property does not change.

For more information, see the **Fill Pattern PropertyManager** topic in the SOLIDWORKS Design user-interface help.

See [Accessing Selections that Define Features](#) for additional details.

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

## FeaturesToPatternType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : FeaturesToPatternType Property (IFillPatternFeatureData)

Gets or sets the type of object to pattern in this fill pattern feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

```
Property FeaturesToPatternType As System.Integer
```

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Integer
```

```
instance.FeaturesToPatternType = value
```

```
value = instance.FeaturesToPatternType
```

### C#

```
System.int FeaturesToPatternType {get; set;}
```

### C++/CLI

```
property System.int FeaturesToPatternType {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Type of object to pattern as defined in [swFeaturesToPatternType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::FeaturesToPatternType](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

### If this property is set to...

### Then call...

swFeaturesToPatternType\_e.swFeaturesToPatternCreateSeedCut  
swFeaturesToPatternType\_e.swFeaturesToPatternSelectedFeatures

[IFillPatternFeatureData::CreateSeedCutType](#) to get or set the type of seed cut to pattern

- [IFillPatternFeatureData::PatternFeatureArray](#) to get or set the array of features to pattern
- [IFillPatternFeatureData::PatternFaceArray](#) to get or set the array of faces to pattern
- [IFillPatternFeatureData::PatternBodyArray](#) to get or set the array of bodies to pattern

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## GeometryPattern Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : GeometryPattern Property (IFillPatternFeatureData)

Gets or sets whether to create this fill pattern using an exact copy of the seed feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

Property GeometryPattern As System.Boolean

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Boolean

instance.GeometryPattern = value
value = instance.GeometryPattern
```

### C#

```
System.bool GeometryPattern {get; set;}
```

### C++/CLI

```
property System.bool GeometryPattern {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to create the pattern using an exact copy of the seed feature, false to solve individual instances of the seed feature

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::GeometryPattern](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if [IFillPatternFeatureData::PatternElement](#) is [swPatternElementSelection\\_e.swFeatureFaces](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## InnerRadius Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : InnerRadius Property (IFillPatternFeatureData)

Gets or sets the radius of the circle that inscribes the polygon-cut seed instance of this fill pattern feature.

## Visual Basic Syntax

### Visual Basic (Declaration)

Property InnerRadius As System.Double

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Double

instance.InnerRadius = value

value = instance.InnerRadius
```

C#

```
System.double InnerRadius {get; set;}
```

C++/CLI

```
property System.double InnerRadius {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Radius of circle inscribing the polygon-cut seed instance

#### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData.InnerRadius](#).

#### Example

See the [IFillPatternFeatureData](#) example.

#### Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e.swFeaturesToPatternCreateSeedCut](#)
- [IFillPatternFeatureData::CreateSeedCutType](#) = [swCreateSeedCutType\\_e.swCreateSeedCutPolygon](#)

#### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

#### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## InstanceSpacing Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : InstanceSpacing Property (IFillPatternFeatureData)

Gets or sets the distance between the pattern instance centers of this fill pattern feature.

#### .NET Syntax

Visual Basic (Declaration)

```
Property InstanceSpacing As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Double

instance.InstanceSpacing = value
```

```
value = instance.InstanceSpacing
```

C#

```
System.double InstanceSpacing {get; set;}
```

C++/CLI

```
property System.double InstanceSpacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Distance between pattern instance centers

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::InstanceSpacing](#).

### Example

See the [IFillPatternFeatureData](#) example.

### Remarks

| This property is.... | If...                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Always valid         | <a href="#">IFillPatternFeatureData::PatternLayoutType</a> = <a href="#">swPatternLayoutType_e.swPatternLayoutPerforation</a>                                                                                                                                                                                                                                                                                                                                                                                                         |
| Only valid           | Both are true:<br><br><a href="#">IFillPatternFeatureData::LayoutSpacingType</a> = <a href="#">swPatternLayoutSpacingType_e.swPatternLayoutTargetSpacing</a><br><br>And:<br><br><a href="#">IFillPatternFeatureData::PatternLayoutType</a> is set to one of the following: <ul style="list-style-type: none"><li>• <a href="#">swPatternLayoutType_e.swPatternLayoutCircle</a></li><li>• <a href="#">swPatternLayoutType_e.swPatternLayoutSquare</a></li><li>• <a href="#">swPatternLayoutType_e.swPatternLayoutPolygon</a></li></ul> |

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

[SOLIDWORKS API Help](#)

[Send comments on this topic.](#)

## LayoutSpacingType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : LayoutSpacingType Property (IFillPatternFeatureData)

Gets or sets the type of spacing between the instances in the layout of this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property LayoutSpacingType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Integer
```

```
instance.LayoutSpacingType = value  
value = instance.LayoutSpacingType
```

C#

```
System.int LayoutSpacingType {get; set;}
```

C++/CLI

```
property System.int LayoutSpacingType {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Pattern layout spacing type as defined in [swPatternLayoutSpacingType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::LayoutSpacingType](#).

### Example

See the [IFillPatternFeatureData](#) example.

### Remarks

#### If this property is set to...

Then also set...

swPatternLayoutSpacingType\_e.swPatternLayoutInstances

[IFillPatternFeatureData::NoOfInstances](#); [IFillPatternFeatureData::InstanceSpacing](#) is not valid

swPatternLayoutSpacingType\_e.swPatternLayoutTargetSpacing

[IFillPatternFeatureData::InstanceSpacing](#); [IFillPatternFeatureData::NoOfInstances](#) is not valid

### See Also

[IFillPatternFeatureData Interface](#)

[IFillPatternFeatureData Members](#)

[IFillPatternFeatureData::PatternLayoutType Property \(\)](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## LoopSpacing Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : LoopSpacing Property (IFillPatternFeatureData)

Gets or sets the distance between loops of pattern instances in this fill pattern feature.

### Visual Basic .NET Syntax

Visual Basic (Declaration)

```
Property LoopSpacing As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Double  
  
instance.LoopSpacing = value
```

```
value = instance.LoopSpacing
```

C#

```
System.double LoopSpacing {get; set;}
```

C++/CLI

```
property System.double LoopSpacing {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Distance between loops of pattern instances

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::LoopSpacing](#).

### Example

See the [IFillPatternFeatureData](#) example.

### Remarks

All fill patterns except those with perforated layouts are created by positioning a pattern seed instance within the fill boundary and copying the pattern in concentric loops about the seed instance. This property sets the spacing between the centers of instances in adjacent pattern loops.

This property is valid only if [IFillPatternFeatureData::PatternLayoutType](#) is set to one of the following:

- [swPatternLayoutType\\_e.swPatternLayoutCircular](#)
- [swPatternLayoutType\\_e.swPatternLayoutPolygon](#)
- [swPatternLayoutType\\_e.swPatternLayoutSquare](#)

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Margins Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : Margins Property (IFillPatternFeatureData)

Gets or sets the distance between the fill boundary and the outermost pattern instance in the layout of this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property Margins As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Double  
  
instance.Margins = value  
  
value = instance.Margins
```

C#

```
System.double Margins {get; set;}
```

C++/CLI

```
property System.double Margins {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Distance between the fill boundary and the outermost pattern instance in the layout; 0 for no margin

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::Margins](#).

### Example

See the [IFillPatternFeatureData](#) example.

### See Also

[IFillPatternFeatureData Interface](#)

[IFillPatternFeatureData Members](#)

[IFillPatternFeatureData::PatternLayoutType Property\(\)](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## NoOfInstances Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : NoOfInstances Property (IFillPatternFeatureData)

Gets or sets the number of instances per loop or side of the layout of this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property NoOfInstances As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Integer
```

```
instance.NoOfInstances = value
```

```
value = instance.NoOfInstances
```

C#

```
System.int NoOfInstances {get; set;}
```

C++/CLI

```
property System.int NoOfInstances {  
    System.int get();  
    void set ( & System.int value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Number of instances per loop or side of the pattern layout

## Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::NoOfInstances](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- And:
- [IFillPatternFeatureData::LayoutSpacingType](#) = [swPatternLayoutSpacingType\\_e](#).swPatternLayoutInstances
  - [IFillPatternFeatureData::PatternLayoutType](#) is set to one of the following:
    - [swPatternLayoutType\\_e](#).swPatternLayoutCircular
    - [swPatternLayoutType\\_e](#).swPatternLayoutPolygon
    - [swPatternLayoutType\\_e](#).swPatternLayoutSquare

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## OuterRadius Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : OuterRadius Property (IFillPatternFeatureData)

Gets or sets the radius of the circle that circumscribes the polygon-cut seed instance of this fill pattern feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Property OuterRadius As System.Double
```

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Double

instance.OuterRadius = value

value = instance.OuterRadius
```

### C#

```
System.double OuterRadius {get; set;}
```

### C++/CLI

```
property System.double OuterRadius {
    System.double get();
    void set ( & System.double value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

## Property Value

Radius of circle circumscribing the polygon-cut seed instance

## Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData.OuterRadius](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut
- [IFillPatternFeatureData::CreateSeedCutType](#) = [swCreateSeedCutType\\_e](#).swCreateSeedCutPolygon

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternBodyArray Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternBodyArray Property (IFillPatternFeatureData)

Gets or sets the array of bodies to pattern in this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property PatternBodyArray As System.Object

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternBodyArray = value
```

```
value = instance.PatternBodyArray
```

C#

```
System.object PatternBodyArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternBodyArray {  
    System.Object^ get();  
    void set ( &    System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Array of [IBody2](#)s to pattern

## Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::PatternBodyArray](#).

## Remarks

This property is valid only if [IFillPatternFeatureData::PatternElement](#) is [swPatternElementSelection\\_e.swBodiesToPattern](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternDirection Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternDirection Property (IFillPatternFeatureData)

Gets or sets the direction reference for the layout of this fill pattern feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Property PatternDirection As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternDirection = value
```

```
value = instance.PatternDirection
```

### C#

```
System.Object PatternDirection {get; set;}
```

### C++/CLI

```
property System.Object^ PatternDirection {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Direction reference for the pattern layout

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternDirection](#).

## Remarks

Before calling this property, call [IFillPatternFeatureData::PatternDirectionType](#) to get the type of pattern direction reference returned by this property. If this property is not specified, SOLIDWORKS chooses an appropriate reference for the specified [IFillPatternFeatureData::PatternLayoutType](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternDirectionType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternDirectionType Property (IFillPatternFeatureData)

Gets the type of pattern direction reference returned by [IFillPatternFeatureData::PatternDirection](#) for this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

ReadOnly Property PatternDirectionType As System.Integer

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.PatternDirectionType
```

C#

```
System.int PatternDirectionType {get;}
```

C++/CLI

```
property System.int PatternDirectionType {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Type of pattern direction reference as defined in [swSelectType\\_e](#)

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternDirectionType](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternElement Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternElement Property (IFillPatternFeatureData)

Gets or sets the pattern element selection of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property PatternElement As System.Integer

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Integer

instance.PatternElement = value

value = instance.PatternElement
```

C#

```
System.int PatternElement {get; set;}
```

C++/CLI

```
property System.int PatternElement {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Pattern element selection as defined in [swPatternElementSelection\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternElement](#).

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2015 FCS, Revision Number 23.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternFaceArray Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternFaceArray Property (IFillPatternFeatureData)

Gets or sets the array of faces to pattern in this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property PatternFaceArray As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Object

instance.PatternFaceArray = value

value = instance.PatternFaceArray
```

C#

```
System.object PatternFaceArray {get; set;}
```

C++/CLI

```
property System.Object^ PatternFaceArray {
    System.Object^ get();
```

```
void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of [IFace2](#)s to pattern

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternFaceArray](#).

### Remarks

This property is valid only if [IFillPatternFeatureData::PatternElement](#) is [swPatternElementSelection\\_e](#).swFeatureFaces.

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternFeatureArray Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternFeatureArray Property (IFillPatternFeatureData)

Gets or sets the array of features to pattern in this fill pattern feature.

### .NET Syntax

#### Visual Basic (Declaration)

```
Property PatternFeatureArray As System.Object
```

#### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object
```

```
instance.PatternFeatureArray = value
```

```
value = instance.PatternFeatureArray
```

#### C#

```
System.object PatternFeatureArray {get; set;}
```

#### C++/CLI

```
property System.Object^ PatternFeatureArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of [IFeatures](#) to pattern

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternFeatureArray](#).

## Remarks

This property is valid only if [IFillPatternFeatureData::PatternElement](#) is [swPatternElementSelection\\_e.swFeatureFaces](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternFillBoundaryArray Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternFillBoundaryArray Property (IFillPatternFeatureData)

Gets or sets the array of objects representing the boundary of this fill pattern feature.

## .NET Syntax

### Visual Basic (Declaration)

```
Property PatternFillBoundaryArray As System.Object
```

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object  
  
instance.PatternFillBoundaryArray = value  
  
value = instance.PatternFillBoundaryArray
```

### C#

```
System.Object PatternFillBoundaryArray {get; set;}
```

### C++/CLI

```
property System.Object^ PatternFillBoundaryArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Array of objects representing the fill boundary

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternFillBoundaryArray](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternLayoutPolygonSides Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternLayoutPolygonSides Property (IFillPatternFeatureData)

Gets or sets the number of sides in the polygonal layout of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property PatternLayoutPolygonSides As System.Integer

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Integer

instance.PatternLayoutPolygonSides = value
value = instance.PatternLayoutPolygonSides
```

C#

```
System.int PatternLayoutPolygonSides {get; set;}
```

C++/CLI

```
property System.int PatternLayoutPolygonSides {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Number of sides in the polygonal layout of this fill pattern

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternLayoutPolygonSides](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if [IFillPatternFeatureData::PatternLayoutType](#) = [swPatternLayoutType\\_e.swPatternLayoutPolygon](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## PatternLayoutType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PatternLayoutType Property (IFillPatternFeatureData)

Gets or sets the layout of the pattern instances within the fill boundary of this fill pattern feature.

## .NET Syntax

```
Property PatternLayoutType As System.Integer
```



```
Dim instance As IFillPatternFeatureData
Dim value As System.Integer

instance.PatternLayoutType = value

value = instance.PatternLayoutType
```

```
System.int PatternLayoutType {get; set;}
```

```
property System.int PatternLayoutType {
    System.int get();
    void set ( & System.int value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Fill pattern layout type as defined in [swPatternLayoutType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PatternLayoutType](#).

### Example

See the [IFillPatternFeatureData](#) example.

### Remarks

After specifying this property, specify the following:

- [IFillPatternFeatureData::Margins](#)
- [IFillPatternFeatureData::PatternDirection](#)
- [IFillPatternFeatureData::GeometryPattern](#)
- [IFillPatternFeatureData::PropagateVisualProperty](#)
- [IFillPatternFeatureData::FeaturesToPatternType](#)

Specify other properties as required by the settings above.

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

## PropagateVisualProperty Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : PropagateVisualProperty Property (IFillPatternFeatureData)

Gets or sets whether to propagate visual properties (e.g., colors, textures, cosmetic thread data, etc.) to all fill pattern instances in this fill pattern feature.

### .NET Syntax

```
Property PropagateVisualProperty As System.Boolean
```

```
Dim instance As IFillPatternFeatureData
Dim value As System.Boolean

instance.PropagateVisualProperty = value
value = instance.PropagateVisualProperty
```

C#

```
System.bool PropagateVisualProperty {get; set;}
```

C++/CLI

```
property System.bool PropagateVisualProperty {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

True to propagate visual properties, false to not

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::PropagateVisualProperty](#).

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments](#) on this topic.

## ReleaseSelectionAccess Method (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : ReleaseSelectionAccess Method (IFillPatternFeatureData)

Releases access to selections that define this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Sub ReleaseSelectionAccess()
```

Visual Basic (Usage)

```
Dim instance As IFillPatternFeatureData

instance.ReleaseSelectionAccess()
```

C#

```
void ReleaseSelectionAccess()
```

C++/CLI

```
void ReleaseSelectionAccess();
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::ReleaseSelectionAccess](#).

## Remarks

[IFillPatternFeatureData::AccessSelections](#) puts the model into a rollback state to allow access to the selections that define the fill pattern feature.

Use this method to restore the rollback state if you did not modify the fill pattern feature, or use [IFeature::ModifyDefinition](#) or [IFeature::IModifyDefinition2](#) if you did modify the fill pattern feature.

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## Rotation Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : Rotation Property (IFillPatternFeatureData)

Gets or sets the counterclockwise rotation of the seed instance of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property Rotation As System.Double

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Double
```

```
instance.Rotation = value
```

```
value = instance.Rotation
```

C#

```
System.double Rotation {get; set;}
```

C++/CLI

```
property System.double Rotation {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

Counterclockwise rotation of the seed instance

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::Rotation](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if both are true:

- [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut

And:

- [IFillPatternFeatureData::CreateSeedCutType](#) is set to one of the following:
  - [swCreateSeedCutType\\_e](#).swCreateSeedCutDiamond
  - [swCreateSeedCutType\\_e](#).swCreateSeedCutPolygon
  - [swCreateSeedCutType\\_e](#).swCreateSeedCutSquare

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SeedCutFlipShapeDirection Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : SeedCutFlipShapeDirection Property (IFillPatternFeatureData)

Gets or sets whether to reverse the direction of the seed instance cut of this fill pattern feature.

## .NET Syntax

Visual Basic (Declaration)

Property SeedCutFlipShapeDirection As System.Boolean

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData
Dim value As System.Boolean
```

```
instance.SeedCutFlipShapeDirection = value
```

```
value = instance.SeedCutFlipShapeDirection
```

C#

```
System.bool SeedCutFlipShapeDirection {get; set;}
```

C++/CLI

```
property System.bool SeedCutFlipShapeDirection {
    System.bool get();
    void set ( & System.bool value);
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

True to reverse the direction of the seed instance cut, false to not

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::SeedCutFlipShapeDirection](#).

## Example

See the [IFillPatternFeatureData](#) example.

## Remarks

This property is valid only if [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e](#).swFeaturesToPatternCreateSeedCut.

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)  
[IFillPatternFeatureData::CreateSeedCutType Property\(\)](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SeedFeatureCenter Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : SeedFeatureCenter Property (IFillPatternFeatureData)

Gets or sets the vertex or sketch point where to place the center of the seed cut feature of this fill pattern feature.

## .NET Syntax

### Visual Basic (Declaration)

Property SeedFeatureCenter As System.Object

### Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object
```

```
instance.SeedFeatureCenter = value
```

```
value = instance.SeedFeatureCenter
```

### C#

```
System.object SeedFeatureCenter {get; set;}
```

### C++/CLI

```
property System.Object^ SeedFeatureCenter {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

### Property Value

[Vertex](#) or [sketch point](#) where to place the center of the seed cut feature

## Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::SeedFeatureCenter](#).

## Remarks

This property is valid only if [IFillPatternFeatureData::FeaturesToPatternType](#) = [swFeaturesToPatternType\\_e.swFeaturesToPatternCreateSeedCut](#).

## See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)  
[IFillPatternFeatureData::SeedFeatureCenterType Property\(\)](#)

## Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SeedFeatureCenterType Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : SeedFeatureCenterType Property (IFillPatternFeatureData)

Gets or sets the type of entity where to place the center of the seed cut feature for this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
ReadOnly Property SeedFeatureCenterType As System.Integer
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Integer
```

```
value = instance.SeedFeatureCenterType
```

C#

```
System.int SeedFeatureCenterType {get;}
```

C++/CLI

```
property System.int SeedFeatureCenterType {  
    System.int get();  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Type of entity as defined in [swSelectType\\_e](#)

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::SeedFeatureCenterType](#).

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)  
[IFillPatternFeatureData::SeedFeatureCenter Property\(\)](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## SkippedItemArray Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : SkippedItemArray Property (IFillPatternFeatureData)

Gets or sets the pattern instances to skip in this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property SkippedItemArray As System.Object
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Object
```

```
instance.SkippedItemArray = value
```

```
value = instance.SkippedItemArray
```

C#

```
System.Object SkippedItemArray {get; set;}
```

C++/CLI

```
property System.Object^ SkippedItemArray {  
    System.Object^ get();  
    void set ( & System.Object^ value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Array of pattern instances to skip (see **Remarks**)

### Visual Basic for Applications (VBA) Syntax

See [FillPatternFeatureData::SkippedItemArray](#).

### Remarks

The array is linear and zero-based and contains the (row\_id, column\_id) of each pattern instance to skip.

For example,

```
[row_id_instance1, column_id_instance1, row_id_instance2, column_id_instance2, ...]
```

### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0

SOLIDWORKS API Help

[Send comments on this topic.](#)

## StaggerAngle Property (IFillPatternFeatureData)

[See Also](#)



[SolidWorks.Interop.sldworks Namespace](#) > [IFillPatternFeatureData Interface](#) : StaggerAngle Property (IFillPatternFeatureData)

Gets or sets the angle by which to stagger the rows of pattern instances in the perforation layout of this fill pattern feature.

### .NET Syntax

Visual Basic (Declaration)

```
Property StaggerAngle As System.Double
```

Visual Basic (Usage)



```
Dim instance As IFillPatternFeatureData  
Dim value As System.Double
```

```
instance.StaggerAngle = value
```

```
value = instance.StaggerAngle
```

C#

```
System.double StaggerAngle {get; set;}
```

C++/CLI

```
property System.double StaggerAngle {  
    System.double get();  
    void set ( & System.double value);  
}
```

**NOTE:** See [Differences Between Unmanaged C++ and C++/CLI Code](#).

#### Property Value

Stagger angle for perforation layout

#### Visual Basic for Applications (VBA) Syntax

See [IFillPatternFeatureData::StaggerAngle](#).

#### Example

See the [IFillPatternFeatureData](#) example.

#### Remarks

This property is valid only if [IFillPatternFeatureData::PatternLayoutType](#) = [swPatternLayoutType\\_e.swPatternLayoutPerforation](#)

#### See Also

[IFillPatternFeatureData Interface](#)  
[IFillPatternFeatureData Members](#)

#### Availability

SOLIDWORKS 2014 FCS, Revision Number 22.0